

Trace Powers versus Hilbert–Schmidt Mass in Nonnormal Cloud Complements

Liang Wang

July 2026

Abstract

RH-229 found that the whole-complement Frobenius estimate grows roughly like σ^{-1} and cannot prove a normal relative determinant family. We show that this failure is not a determinant obstruction. The N -dimensional nilpotent shift has $\|S_N\|_2^2 = N - 1$, while $\text{tr}(S_N^n) = 0$ for every $n \geq 1$ and $\det_2(I - zS_N) \equiv 1$. Nonnormal singular-value mass can therefore diverge without contributing any spectral determinant mass.

For each RH-222 endpoint we compute the exact finite trace of the squared scaled matrix and subtract the Perron, parity, and selected-cloud squares. The maximum complement second-trace modulus is 0.12952, whereas the inherited Hilbert–Schmidt squared upper bound reaches 308.75. The largest ratio is 2.36×10^6 . Hence the Frobenius route is too strong; the next meaningful target is a uniform envelope for the cloud-extracted power traces.

1 Two different notions of complement size

The regularized determinant has logarithmic germ

$$\log \det_2(I - zB) = - \sum_{n \geq 2} \frac{z^n}{n} \text{tr}(B^n).$$

An ideal-norm argument controls these traces through $|\text{tr}(B^n)| \leq \|B^n\|_1$, for example $\|B^2\|_1 \leq \|B\|_2^2$ [1]. The inequality is sufficient but can be extremely wasteful for nonnormal operators.

RH-229 measured precisely this sufficient budget and found it divergent [2]. The present question is whether the divergence is spectral or merely singular-value mass.

2 Exact nilpotent separation

Theorem 2.1 (Divergent Hilbert–Schmidt mass with trivial determinant). *Let $S_N \in \mathbb{C}^{N \times N}$ be the shift with ones on the first superdiagonal and zero elsewhere. Then*

$$\|S_N\|_2^2 = N - 1, \quad \text{tr}(S_N^n) = 0 \quad (n \geq 1), \quad \det_2(I - zS_N) = 1.$$

In particular $\|S_N\|_2 \rightarrow \infty$ while the regularized determinant is identically constant.

Proof. There are $N - 1$ unit entries, proving the Hilbert–Schmidt identity. The matrix is strictly upper triangular, as is every positive power, so every power trace vanishes. All eigenvalues are zero, hence the canonical regularized product equals one. $\square$

This example is maximally nonnormal, but that is exactly its purpose: no logical implication from a divergent Hilbert–Schmidt norm to determinant divergence can hold without additional structure.

Theorem 2.2 (Nontrivial determinant with an invisible HS sector). *Let B be any fixed finite matrix and let $a_N \in \mathbb{C}$. Define*

$$A_N = B \oplus a_N S_N.$$

Then, for every $n \geq 1$,

$$\mathrm{tr}(A_N^n) = \mathrm{tr}(B^n), \quad \det_2(I - zA_N) = \det_2(I - zB),$$

whereas

$$\|A_N\|_2^2 = \|B\|_2^2 + |a_N|^2(N-1).$$

Thus the Hilbert–Schmidt mass can diverge while an arbitrary fixed nontrivial regularized determinant is preserved exactly.

Proof. Powers and traces split over direct sums. The nilpotent block contributes zero to every power trace and has regularized determinant one. The Hilbert–Schmidt identity is the Pythagorean formula for block direct sums. $\square$

This extension shows that the separation is not tied to the constant determinant one. An arbitrarily complicated fixed spectral factor may coexist with an increasingly large nonnormal singular-value sector that is invisible to all power traces.

3 Second-trace extraction

Let $A_{\sigma,d}$ denote either the fine or Haar-coarse scaled matrix. If p_σ , q_σ , and C_σ denote the Perron root, parity root, and selected cloud, define

$$\tau_{2,\sigma} = \mathrm{tr}(A_{\sigma,d}^2) - p_\sigma^2 - q_\sigma^2 - \sum_{\lambda \in C_\sigma} \lambda^2.$$

For a sparse real matrix,

$$\mathrm{tr}(A^2) = \sum_{i,j} A_{ij}A_{ji},$$

so this coefficient can be computed without diagonalizing the unresolved matrix.

Batch statistic	value
Endpoint count	32
Maximum $ \tau_{2,\sigma} $	0.1295197
Minimum $ \tau_{2,\sigma} $	3.8778×10^{-5}
Maximum complement HS-squared upper	308.7521
Maximum HS-squared/ $ \tau_2 $ ratio	2.3619×10^6

Table 1: Spectral trace versus singular-value budget.

The full scaled Frobenius mass grows strongly as the mesh resolves the narrow kernel. By contrast, $\mathrm{tr}(A^2)$ itself remains near 1.633 at the finest scales, and the selected cloud removes nearly all of the remaining spectral second moment.

4 Cancellation efficiency in the archived family

It is useful to record the dimensionless ratio

$$\mathcal{E}_{2,\sigma} = \frac{\text{complement HS-squared upper}}{|\tau_{2,\sigma}|}.$$

This is not a condition number and becomes infinite when the denominator vanishes. It measures only how much larger the cancellation-blind sufficient budget is than the signed spectral coefficient that actually enters $\log \det_2$ at order two. Across the batch it ranges from ordinary two-digit values at the coarsest endpoints to 2.36×10^6 . The sign of τ_2 also changes with noise and channel, confirming that phase and nonnormal cancellation, rather than positive mass, govern this coefficient.

The sparse identity $\text{tr}(A^2) = \sum_{i,j} A_{ij}A_{ji}$ avoids an unresolved eigendecomposition, but subtraction can still lose relative accuracy when τ_2 is tiny. Accordingly the archive treats the values as frozen double-precision observations. A theorem would require interval control of the full trace and the extracted cloud sum, or an analytic trace formula in which their cancellation is performed before numerical evaluation.

5 Why one trace coefficient is not enough

The second coefficient is the first nontrivial term of the regularized logarithm, so its smallness is necessary for convergence to a normalized limit with vanishing quadratic coefficient. It is not sufficient for a normal determinant family. Higher traces may remain large even when $\tau_2 = 0$, and no bound on τ_n for $n \geq 3$ follows from signed cancellation at order two.

There are therefore two logically separate lessons:

1. failure of a Hilbert–Schmidt bound does not close the determinant route;
2. success at one or finitely many trace orders does not close it either.

The useful replacement is an order-uniform, noise-uniform trace envelope, not merely a smaller norm surrogate.

6 What the separation buys

The nilpotent theorem and the finite audit jointly show that the route

$$\text{uniform HS complement} \implies \text{normal relative determinant}$$

is sufficient but not necessary. It is therefore legitimate to replace the failed norm gate by a direct trace gate.

This replacement has a cost. One must control every $n \geq 2$, not merely $n = 2$. RH-236 computes orders through twelve and RH-240 formulates an all-order geometric criterion. Until such a bound is proved, no locally uniform determinant family follows.

No claim about a Hilbert–Polya operator, zeta zeros, or the Riemann Hypothesis is made.

References

- [1] Barry Simon. *Trace Ideals and Their Applications*. American Mathematical Society, second edition, 2005.
- [2] Liang Wang. The nonnormal frobenius tail-budget barrier. 2026. RH-229 preprint.